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Information processing system, setting change method, and non-transitory recording medium

Abstract

An information processing system, a setting change method, and a non-transitory recording medium. The information processing system registers a plurality of access accounts for managing the plurality of information processing apparatuses, transmits, in response to receiving an acquisition request from a particular information processing apparatus of the plurality of information processing apparatuses, the plurality of access accounts to the particular information processing apparatus, determines whether the plurality of access accounts satisfy a requisite based on authority information set in the particular information processing apparatus and stores one of the plurality of access accounts that satisfies the requisite.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is based on and claims priority pursuant to 35 U.S.C. § 119 (a) to Japanese Patent Application No. 2022-030233, filed on Feb. 28, 2022, and Japanese Patent Application No. 2022-192445, filed on Nov. 30, 2022, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

(2) The present disclosure relates to an information processing system, a setting change method, and a non-transitory recording medium.

Background Art

(3) Device management apparatuses for managing a plurality of information processing apparatuses have been developed in such a way that the devices are managed from a server dedicated to the device management apparatus residing in a customer network or personal computer (PC) software installed on a PC of a customer user. This configuration is hereinafter referred to as an on-premises device management apparatus.

(4) In addition, development and popularization of cloud services implemented on the internet have progressed, and development of device management apparatuses managing devices from a cloud server is also progressing. This configuration is hereinafter referred to as a cloud device management apparatus.

SUMMARY

(5) Embodiments of the present disclosure describe an information processing system, a setting change method, and a non-transitory recording medium. The information processing system registers a plurality of access accounts for managing the plurality of information processing apparatuses, transmits, in response to receiving an acquisition request from a particular information processing apparatus of the plurality of information processing apparatuses, the plurality of access accounts to the particular information processing apparatus, determines whether the plurality of access accounts satisfy a requisite based on authority information set in the particular information processing apparatus and stores one of the plurality of access accounts that satisfies the requisite.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

(2) FIG. **1** is a block diagram illustrating a configuration of an information processing system according to embodiments of the present disclosure;

(3) FIG. **2** is a block diagram illustrating a hardware configuration of a multifunction peripheral (MFP), which is an example of an information processing apparatus;

(4) FIG. **3** is a block diagram illustrating a hardware configuration of a PC (server);

(5) FIG. **4** is a block diagram illustrating a functional configuration of the information processing system;

(6) FIG. **5** is a diagram illustrating an example of a user interface (UI) for setting an access account;

(7) FIG. **6** is a diagram illustrating an example of the UI for creating a new access account;

(8) FIG. **7** is a diagram illustrating an example of the UI for deleting the access account;

(9) FIG. **8** is a sequence diagram illustrating a setting change process for the information processing apparatus; and

(10) FIG. **9** is a sequence diagram illustrating an access account acquisition process of the

information processing apparatus.

(11) The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

(12) In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

(13) Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(14) Embodiments of an information processing system, a device management apparatus, an information processing apparatus, a setting change method, and a non-transitory recording medium are described in detail below with reference to accompanying drawings.

(15) FIG. **1** is a block diagram illustrating a configuration of an information processing system **100** according to the present embodiment.

(16) As illustrated in FIG. **1**, an information processing system **100** includes a plurality of information processing apparatuses **10**, a device management apparatus **20**, and an administrator terminal **40** which is a PC.

(17) The information processing apparatus **10** is, for example, an image forming apparatus such as a multifunction peripheral (MFP) including at least two functions out of a copy function, a print function, a scan function, and a facsimile function.

(18) The device management apparatus **20** is an information system for device management for managing the information processing apparatus **10** through the firewall in a customer network **30**. The device management apparatus **20** is implemented by one or more PCs (servers) **5** provided on a cloud service implemented on the internet.

(19) A hardware configuration of an MFP **9**, which is an example of the information processing apparatus **10** is described in the following.

(20) FIG. **2** is a block diagram illustrating a hardware configuration of the MFP **9**, which is an example of the information processing apparatus **10**.

(21) As illustrated in FIG. **2**, the MFP **9**, which is the information processing apparatus **10**, includes a controller **910** included in a device main body, a short-range communication circuit **920**, an engine controller **930**, a control panel **940**, and a network interface (I/F) **950**.

(22) The controller **910** includes a CPU **901** as a main processor, a system memory (MEM-P) **902**, a north bridge (NB) **903**, a south bridge (SB) **904**, an Application Specific Integrated Circuit (ASIC) **906**, a local memory (MEM-C) **907** as a storage unit, a hard disk drive (HDD) controller **908**, and a hard disk (HD) **909** as another storage unit. The NB **903** and the ASIC **906** are connected through an Accelerated Graphics Port (AGP) bus **921**.

(23) The CPU **901** is a processor that performs overall control of the MFP **9**. The NB **903** connects the CPU **901** with the MEM-P **902**, SB **904**, and AGP bus **921**. The NB **903** includes a memory controller for controlling reading or writing of various data with respect to the MEM-P **902**, a Peripheral Component Interconnect (PCI) master, and an AGP target.

(24) The MEM-P **902** includes a read only memory (ROM) **902***a* as a memory that stores program and data for implementing various functions of the controller **910**. The MEM-P **902** further includes a RAM **902***b* as a memory that deploys the program and data, or as a drawing memory that stores drawing data for printing. The program stored in the ROM **902***a* may be stored in any computer-readable storage medium, such as a Compact Disc-Read Only Memory (CD-ROM),

Compact Disc-Recordable (CD-R), or Digital Versatile Disc (DVD), in a file format installable or executable by the computer for distribution.
(25) The SB **904** is a bridge for connecting the NB **903** with a peripheral component interconnect (PCI) device or a peripheral device. The ASIC **906** is an integrated circuit (IC) dedicated to an image processing use, and connects the AGP bus **921**, a PCI bus **922**, the HDD controller **908**, and the MEM-C **907**. This ASIC **906** includes a PCI target and AGP master, an arbiter (ARB) that forms the core of the ASIC **906**, a memory controller that controls the MEM-C **907**, and multiple Direct Memory Access Controllers (DMACs) that rotate image data using hardware logic, and a PCI unit that transfers data between the scanner **931** and the printer **932** through the PCI bus **922**. The ASIC **906** may be connected to a Universal Serial Bus (USB) interface, or an Institute of Electrical and Electronics Engineers **1394** (IEEE1394) interface.
(26) The MEM-C **907** is a local memory used as a buffer for image data to be copied or code image. The HD **909** is a storage that stores image data, font data used during printing, and forms. The HDD controller **908** reads or writes various data from or to the HD **909** under control of the CPU **901**. The AGP bus **921** is a bus interface for a graphics accelerator card, which has been proposed to accelerate graphics processing. Through directly accessing the MEM-P **902** by high-throughput, speed of the graphics accelerator card is improved.
(27) The short-range communication circuit **920** is provided with a short-range communication antenna **920***a*. The short-range communication circuit **920** is a communication circuit that communicates in compliance with Near Field Communication (NFC), BLUETOOTH (registered trademark) and the like.
(28) The engine controller **930** includes a scanner **931** and a printer **932**. The control panel **940** includes a display panel **940***a* and an operation panel **940***b*. The display panel **940***a* is implemented by, for example, a touch panel that displays current settings or a selection screen and receives a user input. The operation panel **940***b* includes a numeric keypad that receives set values of various image forming parameters such as image density parameter and a start key that accepts an instruction for starting copying. The controller **910** controls entire operation of the MFP **9**. For example, the controller **910** controls rendering, communication, or user inputs to the control panel **940**. The scanner **931** or the printer **932** includes image processing functions such as error diffusion processing and gamma conversion processing.
(29) In response to an instruction to select a specific application through the control panel **940**, for example, using a mode switch key, the MFP **9** selectively performs a document box function, a copy function, a print function, and a facsimile function. The document box mode is selected when the document box function is selected, the copy mode is selected when the copy function is selected, the print mode is selected when the print function is selected, and the facsimile mode is selected when the facsimile function is selected.
(30) The network I/F **950** is an interface for performing data communication using the customer network **30**. The short-range communication circuit **920** and the network I/F **950** are electrically connected to the ASIC **906** through the PCI bus **922**.
(31) A hardware configuration of the PC (server) **5** for implementing the device management apparatus **20** provided on the cloud service is described in the following.
(32) FIG. **3** is a block diagram illustrating the hardware configuration of the PC (server) **5**. A description is now given of the hardware configuration of the server **5**.
(33) As illustrated in FIG. **3**, the server **5** is implemented by a computer and includes a CPU **501**, a ROM **502**, a RAM **503**, a hard disk (HD) **504**, a hard disk drive (HDD) controller **505**, a display **506**, an external device connection I/F **508**, a network I/F **509**, a bus line **510**, a keyboard **511**, a pointing device **512**, a Digital Versatile Disc-Rewritable (DVD-RW) drive **514**, and a medium I/F **516**.
(34) The CPU **501** controls entire operation of the server **5**. The ROM **502** stores a control program such as an initial program loader (IPL) to boot the CPU **501**. The RAM **503** is used as a work area

for the CPU **501**. The HD **504** stores various data such as programs. The HDD controller **505** controls reading and writing of various data from and to the HD **504** under control of the CPU **501**. The display **506** displays various information such as a cursor, menu, window, character, or image. The external device connection I/F **508** is an interface for connecting various external devices. Examples of the external devices include, but are not limited to, a USB memory and a printer. The network I/F **509** is an interface for performing data communication using the customer network **30**. The bus line **510** is an address bus, a data bus, or the like for electrically connecting each component such as the CPU **501** illustrated in FIG. **3**.
(35) The keyboard **511** is an example of an input device provided with a plurality of keys for allowing a user to input characters, numerals, or various instructions. The pointing device **512** is an example of the input device that allows a user to select or execute a specific instruction, select a target for processing, or move a cursor being displayed. The DVD-RW drive **514** reads and writes various data from and to a DVD-RW **513**, which is an example of a removable storage medium. The removable storage medium is not limited to the DVD-RW and may be a Digital Versatile Disc-Recordable (DVD-R) or the like. The medium I/F **516** controls reading and writing (storing) of data from and to a storage medium **515** such as a flash memory.
(36) A description is now given of functions implemented by the controller **910** of the MFP **9** that is the information processing apparatus **10**, according to the programs stored in the ROM **902***a*, and functions implemented by the CPU **501** of the server **5** implementing the device management apparatus **20** according to the programs stored in the ROM **502** and the HD **504**.
(37) FIG. **4** is a block diagram illustrating a functional configuration of the information processing system **100**. As illustrated in FIG. **4**, the device management apparatus **20** includes a device settings registration unit **21** as an example of a first registration unit, a first communication unit **23** as an example of a transmission unit, and a device settings storage unit **22**. The MFP **9** that is the information processing apparatus **10** includes a device settings processing unit **11** that is an example of a determination unit, a device settings change unit **12**, a device account registration unit **13** that is an example of a second registration unit, a second communication unit **14**, and a storage unit **15**.
(38) The device settings registration unit **21** provides the administrator terminal **40** with a setting change execution screen, which is an input screen for requesting to change the device settings of the information processing apparatus **10**. Further, the device settings registration unit **21** displays a change result of the device settings of the information processing apparatus **10** on the administrator terminal **40** as a device settings confirmation screen. A setting change request for the information processing apparatus **10** is assumed to be received from the administrator terminal **40**.
(39) The device settings storage unit **22** stores the device settings of the information processing apparatus **10** received from the device settings processing unit **11** in an access account storage area for the information processing apparatus. The device settings storage unit **22** manages requests for changing the device settings of the information processing apparatus **10** input from the administrator terminal **40**. Further, the device settings storage unit **22** stores the access account in the device settings storage unit **22** by receiving registration of the access account from the user. The first communication unit **23** transmits an instruction to change the device settings of the information processing apparatus **10** to the device settings processing unit **11**. Also, a transmission request including device settings is received from the second communication unit **14**.
(40) The second communication unit **14** acquires device settings of the information processing apparatus **10** from the device settings change unit **12**.
(41) Also, the second communication unit **14** periodically transmits a transmission request including the device settings of the information processing apparatus **10** to the device settings registration unit **21**. The device settings processing unit **11** executes processing such as changing device settings according to the type of instruction received from the first communication unit **23**. In response to receiving an instruction to change the device settings of the information processing

apparatus **10** from the first communication unit **23**, the device settings processing unit **11** transfers the instruction to the device settings change unit **12**.
(42) The device settings change unit **12** acquires device settings from the controller **910** included in the device main body of the information processing apparatus **10**. The device settings change unit **12** changes device settings managed by the controller **910** included in the device main body of the information processing apparatus **10** based on the change instruction received from the first communication unit **23**. The device account registration unit **13** receives registration of a device account from a user and stores the device account. The storage unit **15** stores the device account registered by the device account registration unit **13** and further stores the access account acquired from the device management apparatus **20**.
(43) In the case a customer owns a plurality of information processing apparatuses **10** and wants to change the device settings of those information processing apparatuses **10**, the customer may wish to apply the same device settings to all the information processing apparatuses **10**. In this case, the customer desires to collectively change the settings of all the information processing apparatuses **10** or a plurality of specific information processing apparatuses **10** with one setting change operation, instead of changing the settings of the information processing apparatuses **10** one by one.
(44) Using a common account for multiple information processing apparatuses **10** may be efficient when changing the device settings of the information processing apparatus **10**, but a plurality of device accounts may be set in the information processing apparatus **10** in proportion to the number of devices and the number of departments.
(45) In order to enable the customer's device administrator to enter the access account settings to be applied to all the information processing apparatuses **10** collectively through the administrator terminal **40**, the device settings registration unit **21** of the device management apparatus **20** displays a setting change execution screen, which is an access account setting UI for defining the setting contents of the access account to be applied to all information processing apparatuses **10** or a plurality of specific information processing apparatuses **10** on the administrator terminal **40**, and causes the customer's device administrator to input a settings change instruction to the information processing apparatus **10** whose setting is to be changed through the administrator terminal **40** using the access account.
(46) FIG. **5** is a diagram illustrating an example of the setting UI of the access account. As illustrated in FIG. **5**, a setting change execution screen, which is the setting UI, includes five items of information, “Priority”, “Profile Name”, “Description”, “User Name”, and “Association with Device”. The priority is information indicating order of determination in the device settings processing unit **11**. The device settings processing unit **11** determines whether the access account satisfies a predetermined requisite in the order indicated by the priority, as described below. The profile name is information for identifying the device settings. The description is any comment entered by the administrator. The user name is information for identifying the administrator, such as a name or an identifier (ID). The association with the device is information related to the information processing apparatus **10** that is stored in association with the access account. Note that the association with the device is configured by an input component such as a setting button. In response to an operation of the setting button by the administrator through the administrator terminal **40**, the device management apparatus **20** provides the administrator terminal **40** with information identifying the information processing apparatus **10** stored in association with the access account. This allows the administrator to confirm or change the information processing apparatus **10** that is stored in association with the access account.
(47) Each information processing apparatus **10** is associated with at least one administrator setting of the information processing apparatus **10**.
(48) Here, the administrator setting is information on at least one of the priority, profile name, description, user name, and association with device. A default administrator setting of the

information processing apparatus **10** is associated immediately after the registration of the information processing apparatus **10**. Changes to the settings of any registered information processing apparatus **10** may be made using the setting button in the “association with device” column. Further, the information is automatically updated through registration by the device settings processing unit **11** of the information processing apparatus **10**, as described below. In the present embodiment, the association registered later becomes effective.
(49) Although the priority can be changed and the order of transmitting the access accounts to the information processing apparatus **10** and determining whether the access accounts satisfy the requisite changes, but the association remains unchanged.
(50) Also, as described above, the administrator settings for a plurality of information processing apparatuses **10** can be registered, but the default administrator is to be set for the information processing apparatus **10**.
(51) Creation and deletion of the access account is described in the following.
(52) FIG. **6** is a diagram illustrating an example of the UI for creating a new access account. The setting change execution screen, which is the UI for creating the new access account, sets the user name and password of the device administrator of the information processing apparatus **10** as illustrated in FIG. **6**. The default administrator settings of the information processing apparatus **10** include the following. Profile name: default (unchangeable) Description: empty User name: admin Password: empty
(53) FIG. **7** is a diagram illustrating an example of the UI for deleting the access account. As illustrated in FIG. **7**, in the setting change execution screen, which is a UI for deletion, by selecting “Delete Device Administrator” in “Other Menu”, the access account for the information processing apparatus **10** whose check box is checked is deleted.
(54) With the above functional configuration, since the device management apparatus **20** transmits the change request as a response to the request from the information processing apparatus **10**, the change request for the settings of the information processing apparatus **10** is delivered to the information processing apparatus **10** placed in the customer network **30** protected by the customer's firewall.
(55) A setting change process for the information processing apparatus **10** is described in the following.
(56) FIG. **8** is a sequence diagram illustrating a setting change process for the information processing apparatus **10**.
(57) The information processing system **100** of the present embodiment initiates communication from the information processing apparatus **10** to the device management apparatus **20** to enable communication through the firewall within the customer network **30**. More specifically, in steps S**1** and S**2**, the device settings processing unit **11** of the information processing apparatus **10** transmits a device settings request to the controller **910** included in the device main body, periodically, for example, at five minute intervals, through the device settings change unit **12** as illustrated in FIG. **8**. In steps S**3** and S**4**, the device settings processing unit **11** of the information processing apparatus **10** receives the device settings information from the controller **910** included in the device main body through the device settings change unit **12**.
(58) That is, the device settings processing unit **11** of the information processing apparatus **10** transmits to the device management apparatus **20** through the second communication unit **14**, a transmission request including current device settings of the own device, periodically, for example, at five minute intervals.
(59) In step S**5**, the device settings storage unit **22** of the device management apparatus **20** receives the device settings from the information processing apparatus **10** through the first communication unit **23**. The device settings may include identification information for identifying the information processing apparatus **10**.
(60) In response to receiving the device settings from the information processing apparatus **10**, the

device settings storage unit **22** of the device management apparatus **20** checks whether the administrator terminal **40** has received a request to change the settings of the information processing apparatus **10**. That is, the administrator terminal **40** transmits a change request to the information processing apparatus **10** by accessing the device management apparatus **20** in advance. The change request includes an item of the device settings to be changed, a setting value after the change, and the identification information of the information processing apparatus **10** to change the device settings. In the case the device settings storage unit **22** of the device management apparatus **20** stores the device settings change request for the information processing apparatus **10**, a response to the communication from the information processing apparatus **10** including the device settings change request is returned to the information processing apparatus **10** through the first communication unit **23** in step S**6**. Note that the device settings storage unit **22** compares the device settings included in the transmission request with the device settings included in the change request, transmits a response in the case the setting value of the setting item is different, and omits the response in the case the setting value of the setting item is the same.

(61) On the other hand, in the case the device settings storage unit **22** of the device management apparatus **20** has not acquired the access account that satisfies the requisite, instead of transmitting the device settings request to the device settings change unit **12**, a notification indicating that there is no account that satisfies the requisite is transmitted to the information processing apparatus **10** in step S**7**.

(62) A process for acquiring the access account of the information processing apparatus **10** is described in detail in the following.

(63) FIG. **9** is a sequence diagram illustrating an access account acquisition process of the information processing apparatus **10**. In step S**11**, the device administrator registers a device account in the device account registration unit **13**, as illustrated in FIG. **9**. A user name, password, and authority information are set for the device account. In step S**12**, the device settings storage unit **22** registers an access account in the device management apparatus **20** through the UI for creating the new access account of the device management apparatus **20** illustrated in FIG. **6**.

(64) Here, the device account is an account registered in the information processing apparatus **10** and the access account is an account registered in the device management apparatus **20**. The access account corresponds to the device account of any one of the information processing apparatuses **10** managed by the device management apparatus **20** or one of the information processing apparatuses **10**.

(65) Note that a plurality of device accounts can be registered in the information processing apparatus **10**. One or more of a plurality of authorities such as device administrator authority, network administrator authority, document administrator authority, and user administrator authority can be registered for each device account. The device administrator authority is the authority of the administrator who manages the information processing apparatus **10**. The network administrator authority is the authority of an administrator who manages the network of the customer environment to which the information processing apparatus **10** is connected. The document manager authority is the authority of an administrator who manages document data processed by the information processing apparatus **10**. The user administrator authority is the authority of an administrator who manages users.

(66) Also, a plurality of access accounts can be registered in the device management apparatus **20**. When registering the access account in the device management apparatus **20**, an access account having at least information identifying the user such as the user name or the user ID matching the device account registered in the information processing apparatus **10** is to be registered. Here, matching of the user names of the device account and the access account may be referred to as the device account and the access account correspond to each other.

(67) Note that when a plurality of device accounts are registered in the information processing apparatus **10**, the device management apparatus **20** can register an access account corresponding to

at least one device account registered in the information processing apparatus **10**. When the device account is registered in each of the plurality of information processing apparatuses **10**, the access account corresponding to at least one device account registered in each of the plurality of information processing apparatuses **10** can be registered.

(68) As illustrated in FIG. **9**, the device settings processing unit **11** of the information processing apparatus **10** transmits an acquisition request requesting all access accounts to the device settings storage unit **22** of the device management apparatus **20** through the second communication unit **14** in step S**13**, and acquires all access accounts from the device settings storage unit **22** in step S**14**. That is, the first communication unit transmits the access accounts to the second communication unit. Here, the access account is an administrator account registered in the information processing apparatus **10** to be managed. In the case the plurality of information processing apparatuses **10** are to be managed, the access account corresponding to the device account registered in each of the plurality of information processing apparatuses **10** are to be registered in advance in the device settings storage unit **22**. In addition to the access account corresponding to the device account pre-registered in the information processing apparatus **10** that is the source of the request, the device settings storage unit **22** transmits the access accounts corresponding to the device accounts registered in other information processing apparatuses **10** to the information processing apparatus **10** through the first communication unit **23**. The device settings storage unit **22** may transmit all access accounts registered in the device settings storage unit **22** to the information processing apparatus **10**, but the transmission to the information processing apparatus **10** may be made in the order indicated by the priority set in the access account. In this case, after transmitting the access account to the information processing apparatus **10**, the device settings storage unit **22** transmits the next highest priority access account to the information processing apparatus **10** in response to a request from the information processing apparatus **10**.

(69) In step S**15**, the device settings processing unit **11** of the information processing apparatus **10** determines whether each of the acquired access accounts satisfies the requisite. Here, the device settings processing unit **11** determines whether the access account acquired from the device management apparatus **20** corresponds to the device account registered in the information processing apparatus **10**. For example, in the case the user name included in the access account matches the user name included in the device account, the device settings processing unit **11** of the information processing apparatus **10** determines that the access account corresponds to the device account. Note that the determination that the access account corresponds to the device account may be made in the case both the user name and password included in the access account match the user name and password included in the device account.

(70) Furthermore, the device settings processing unit **11** determines whether the device account corresponding to the access account has a predetermined authority. For example, in the case all of the device administrator authority, network administrator authority, document administrator authority, and user administrator authority are set for the device account, a determination that the user has the predetermined authority is made. The device settings processing unit **11** determines that the access account satisfies the requisite in the case the device account corresponding to the access account acquired from the device management apparatus **20** is registered in the information processing apparatus **10**, and the device account has the specified authority. Note that the device settings processing unit **11** makes the above determination for each access account acquired from the device management apparatus **20**. Further, the device settings processing unit **11** makes determination of the above-described requisite in the order according to the priority set in the access account. Based on a determination that one of the plurality of access accounts acquired from the device management apparatus **20** satisfies the requisite, the device settings processing unit **11** stops making determination of the remaining access accounts that have not yet been determined.

(71) Thereby, the device settings processing unit **11** specifies one access account among the plurality of access accounts acquired from the device management apparatus **20**.

(72) As soon as the access account satisfying the requisite is found, the device settings processing unit **11** of the information processing apparatus **10** stores the access account in the storage unit **15** in step S**16**. Here, the storage unit **15** may store the device account corresponding to the access account as the access account. In step S**17**, the device settings processing unit **11** of the information processing apparatus **10** transmits a registration request including the device settings of the access account and the identification information of the information processing apparatus **10** to the device settings storage unit **22** of the device management apparatus **20** through the second communication unit **14**.
(73) In step S**18**, the device settings processing unit **11** of the information processing apparatus **10** associates and stores the access account received from the information processing apparatus **10** through the first communication unit **23** and the identification information of the information processing apparatus **10** that is the transmission source. As a result, the identification information of the information processing apparatus **10** is written in the access account storage area for each information processing apparatus **10** of the device management apparatus **20**.
(74) Thereafter, when the device settings processing unit **11** of the information processing apparatus **10** determines that the access account is associated with the identification information of the information processing apparatus **10** in the memory, the access account is used to communicate with the information processing apparatus **10**.
(75) On the other hand, in the case the access account is not stored in the memory, but the history information indicating that the access account acquired from the device management apparatus **20** is stored, the device settings processing unit **11** of the information processing apparatus **10** requests the access account for the information processing apparatus **10** from the device settings storage unit **22** of the device management apparatus **20** in step S**19**, and acquires the access account of the information processing apparatus **10** from the device settings storage unit **22** of the device management apparatus **20** in step S**20**.
(76) In the case an access account is stored in the memory, the device settings processing unit **11** of the information processing apparatus **10** acquires the access count in step S**21**. However, in one example, the authority information set in the device account corresponding to the acquired access account fails to satisfy the prescribed requisite. For example, the authority information of the device account is changed after being stored as an access account in the past. In this case, the processing from step S**13** described above can be executed.
(77) As described above, according to the present embodiment, an account can be set in the information processing apparatus **10** managed by the device management apparatus **20** in the cloud. According to the present embodiment, determination of whether a plurality of different access accounts for each managed device acquired from the device management apparatus **20** satisfies the requisite can be made, the access account that satisfies the requisite is stored in the storage unit **15**, and the stored access account is valid until the application installed in the information processing apparatus **10** or the information processing apparatus is shut down. Note that, according to the present embodiment, even when the information processing apparatus **10** or the application installed in the information processing apparatus **10** is restarted, the access account can be acquired again from the device management apparatus **20**.
(78) A program to be executed by the information processing apparatus **10** or the device management apparatus **20** according to the present embodiment is recorded as a file of an installable format or an executable format on a non-transitory computer-readable recording medium such as a Compact Disc Read-Only Memory (CR-ROM), a flexible disk (FD), a Compact Disc Recordable (CD-R), or a Digital Versatile Disc (DVD).
(79) In another example, the program to be executed by the information processing apparatus **10** or the device management apparatus **20** according to the present embodiment is stored on a computer connected to a network such as the internet and is provided by being downloaded through the network. Also, the program executed by the information processing apparatus **10** or the device

management apparatus **20** of the present embodiment may be provided or distributed through a network such as the internet.

(80) Further, the information processing apparatus **10** or the device management apparatus **20** of the present embodiment may be configured to be incorporated in a ROM or the like in advance and provided.

(81) The program to be executed by the information processing apparatus **10** or the device management apparatus **20** according to the present embodiment is preinstalled in a ROM or the like and provided.

(82) In the above embodiments, an example in which the information processing apparatus **10** of the present disclosure is the MFP including at least two of a copy function, a print function, a scan function, and a facsimile function is described, but the above embodiments may be applied to any image forming apparatuses such as a printer, a scanner, and a facsimile.

(83) Note that the information processing apparatus **10** is not limited to the image forming apparatus as long as the apparatus includes the communication function. The information processing apparatus **10** includes, for example, an output device such as a projector (PJ), an interactive white board (IWB: a white board having an electronic whiteboard function capable of mutual communication), and a digital signage, a head up display (HUD) device, an industrial machine, an imaging device, a sound collecting device, a medical device, a network home appliance, an automobile (connected car), a notebook PC, a mobile phone, a smartphone, a tablet terminal, a game console, a personal digital assistant (PDA), a digital camera, a wearable PC, and a desktop PC.

(84) The apparatuses or devices described in one or more embodiments are just one example of plural computing environments that implement the one or more embodiments disclosed herein.

(85) In some embodiments, the device management apparatus **20** includes multiple computing devices, such as a server cluster. The multiple computing devices are configured to communicate with one another through any type of communication link, including a network, shared memory, etc., and perform the processes disclosed herein. In substantially the same manner, for example, the information processing apparatus **10** includes such multiple computing devices configured to communicate with one another.

(86) Further, the device management apparatus **20** and the information processing apparatus **10** can be configured to share the disclosed processing steps, for example, FIGS. **8** and **9**, in various combinations. For example, a process executed by a given unit may be executed by the information processing apparatus **10**. Similarly, the functions of a given unit can be performed by the information processing apparatus **10**. Also, each element of the device management apparatus **20** and the information processing apparatus **10** may be integrated into one server apparatus, or may be divided into a plurality of apparatuses.

(87) Aspects of the present disclosure are, for example, as follows.

(88) According to a first aspect, an information processing system including a plurality of information processing apparatuses and a device management apparatus that manages the plurality of information processing apparatuses includes, a first registration unit for registering an access account, a transmission unit for transmitting a plurality of access accounts to the information processing apparatus in response to receiving an acquisition request from the information processing apparatus, a determination unit for determining whether the plurality of access accounts satisfy a requisite, and a storage unit for storing one of the plurality of access accounts that satisfies the requisite based on the determination by the determination unit that the requisite is satisfied.

(89) According to a second aspect, in the information processing system of the first aspect, the first registration unit stores the access account in an access account storage area for the information processing apparatus, and in response to receiving an acquisition request from the information processing apparatus, the transmission unit transmits the access account stored in the access account storage area.

(90) According to a third aspect, the information processing system of the first aspect or the second aspect further includes a second registration unit for registering a device account including user identification information, and the determination unit determines that the requisite is satisfied when the user identification information included in the access account matches the user identification information included in the device account.
(91) According to a fourth aspect, in the information processing system of the third aspect, the second registration unit further registers the device account including user authority information, and the determination unit determines that the requisite is satisfied when the user authority information included in the device account corresponding to the access account includes a predetermined authority.
(92) According to a fifth aspect, in the information processing system of any one of the first aspect to the fourth aspect, the determination unit re-acquires the access account from the device management apparatus even when the information processing apparatus or an application installed in the information processing apparatus is restarted.
(93) According to a sixth aspect, in an information processing system including a plurality of information processing apparatuses and a device management apparatus that manages the plurality of information processing apparatuses, the device management apparatus includes a registration unit for registering an access account, a transmission unit for transmitting a plurality of access accounts to the information processing apparatus in response to receiving an acquisition request from the information processing apparatus, and the information processing apparatus includes a determination unit for determining whether the plurality of access accounts satisfy a requisite, and a storage unit for storing one of the plurality of access accounts that satisfies the requisite based on a determination by the determination unit that the requisite is satisfied.
(94) According to a seventh aspect, a device management apparatus that manages a plurality of information processing apparatuses includes, a registration unit for registering an access account, and a transmission unit for transmitting a plurality of access accounts to the information processing apparatus in response to receiving an acquisition request from the information processing apparatus.
(95) According to an eighth aspect, an information processing apparatus managed by a device management apparatus includes, a determination unit for determining whether a plurality of access accounts satisfy a requisite, and a storage unit for storing an access account that satisfies the requisite based on a determination by a determination unit that the requisite is satisfied.
(96) According to a ninth aspect, a setting change method in an information processing system including a plurality of information processing apparatuses and a device management apparatus that manages the plurality of information processing apparatuses includes, a registration step of registering an access account, a transmission step of transmitting a plurality of access accounts to the information processing apparatus in response to receiving an acquisition request from the information processing apparatus, a determination step of determining whether the plurality of access accounts satisfies a requisite, and a storing step of storing the access account that satisfies the requisite based on a determination in the determination step that the requisite is satisfied.
(97) According to a tenth aspect, a program causes a computer for controlling a device management apparatus that manages a plurality of information processing apparatuses to function as, a registration unit for registering an access account, and a transmission unit for transmitting a plurality of access accounts to the information processing apparatus in response to receiving an acquisition request from the information processing apparatus.
(98) According to an eleventh aspect, a program causes a computer for controlling an information processing apparatus managed by a device management apparatus to function as a determination unit for determining whether a plurality of access accounts satisfies a requisite, and storage unit for storing one of the plurality of access accounts that satisfies the requisite based on a determination by a determination unit that the requisite is satisfied.

(99) The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention. Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.
(100) The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, application specific integrated circuits (ASICs), digital signal processors (DSPs), field programmable gate arrays (FPGAs), conventional circuitry and/or combinations thereof which are configured or programmed to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein or otherwise known which is programmed or configured to carry out the recited functionality. When the hardware is a processor which may be considered a type of circuitry, the circuitry, means, or units are a combination of hardware and software, the software being used to configure the hardware and/or processor.

Claims

1. An information processing system comprising a plurality of information processing apparatuses and a device management apparatus that manages the plurality of information processing apparatuses, the device management apparatus including: device management circuitry configured to: register a plurality of access accounts for managing the plurality of information processing apparatuses; and transmit, in response to receiving an acquisition request from a particular information processing apparatus of the plurality of information processing apparatuses, the plurality of access accounts to the particular information processing apparatus, and the particular information processing apparatus including: a memory; information processing circuitry configured to: receive a registration of a device account from a user; store the device account including user identification information in the memory; determine that a requisite is satisfied in a case in which user identification information included in a matching access account of the plurality of access accounts matches the user identification information included in the device account; transmit a registration request to the device management apparatus, wherein the registration request includes the matching access account that satisfies the requisite and includes identification information of the particular information processing apparatus; and store one of the plurality of access accounts that satisfies the requisite, wherein the one of the plurality of access accounts that is stored being the matching access account, wherein the device management circuitry is further configured to: responsive to receipt of the registration request, store and associate the received matching access account with the received identification information of the particular information processing apparatus; and further communicate with the particular information processing apparatus using the stored matching access account associated with the stored identification information of the particular information processing apparatus.
2. The information processing system of claim 1, wherein the device management circuitry is configured to: store the matching access account in an account storage area for the information processing apparatus; and in response to receiving the acquisition request from the information processing apparatus, transmit the matching access account stored in the account storage area.
3. The information processing system of claim 1, wherein the information processing circuitry is further configured to: register a device account including user authority information; and determine that the requisite is satisfied in a case the user authority information included in a device account

corresponding to the matching access account includes a predetermined authority.

4. The information processing system of claim 1, wherein the information processing circuitry is further configured to acquire the matching access account again from the device management apparatus in response to a restart of the information processing apparatus.

5. The information processing system of claim 1, wherein the information processing circuitry is further configured to acquire the matching access account again from the device management apparatus in response to a restart of an application installed in the information processing apparatus.

6. A method for changing setting, executed by an information processing system comprising a plurality of information processing apparatuses and a device management apparatus that manages the plurality of information processing apparatuses, the method including: with the device management apparatus, registering a plurality of access accounts for managing the plurality of information processing apparatuses; with the device management apparatus, transmitting, in response to receiving an acquisition request from a particular information processing apparatus of the plurality of information processing apparatuses, the plurality of access accounts to the particular information processing apparatus; with the particular information processing apparatus, receiving a registration of a device account from a user; with the particular information processing apparatus, storing the device account including user identification information in a memory; with the particular information processing apparatus, determining that a requisite is satisfied in a case in which user identification information included in a matching access account of the plurality of access accounts matches the user identification information included in the device account; with the particular information processing apparatus, transmitting a registration request to the device management apparatus, wherein the registration request includes the matching access account that satisfies the requisite and includes identification information of the particular information processing apparatus; with the particular information processing apparatus, storing one of the plurality of access accounts that satisfies the requisite, wherein the one of the plurality of access accounts that is stored being the matching access account; with the device management apparatus, responding to receipt of the registration request, store and associate the received matching access account with the received identification information of the particular information processing apparatus; and with the device management apparatus, further communicating with the particular information processing apparatus using the stored matching access account associated with the stored identification information of the particular information processing apparatus.